

Week 3 - Day 3

Overview

In this day, I configured Entity Framework Core in the ASP.NET Core Web API project and created the database structure using entities, DbContext, and migrations. The database was created locally through the project environment.

1. Installed EF Core Packages

Added the required Entity Framework Core packages:

- Microsoft.EntityFrameworkCore
- Microsoft.EntityFrameworkCore.Tools
- Microsoft.EntityFrameworkCore.SqlServer

These packages allow the application to:

- Map C# classes to database tables
- Manage database changes using migrations
- Perform database operations using EF Core

2. Created Entity Classes

Created entity classes based on the ERD design.

Entities:

- Book
- Author
- Translator
- Customer
- Order
- OrderItem

Each entity includes:

- Properties representing the data fields
- Navigation properties to represent relationships between entities

Example relationship:

Author - Book is : 1 to many

Translator - book is : 1 to many

3. Created DbContext

Created an application DbContext to manage database communication.

The DbContext contains DbSet properties for each entity:

For example :

```
public DbSet<Book> Books {get;set; }
public DbSet<Author> Authors {get;set; }
public DbSet<Translator> Translators {get;set; }
.
.
.
```

It is responsible for:

- Tracking entities
- Managing relationships
- Applying database change

4. Configured EF Core in the Application

Registered the DbContext inside `Program.cs` and configured the connection to the local database used by the project

5. Created Database Migration

Created the first migration using:

```
dotnet ef migrations add InitialCreate
```

The migration file contains the database creation instructions based on the entity models.

6. Applied Migration

Applied the migration to create the database structure:

```
dotnet ef database update
```

The database tables were generated locally based on the defined entities and relationship

Tools Used

- Entity Framework Core
- ASP.NET Core Web API
- C#
- Visual Studio
- Local Development Database